

1. An investment firm offers its customers municipal bonds that mature after random number T of years. Below is the cumulative distribution function of T . [20 points]

$$F(t) = \begin{cases} 0, & \text{if } t < 1 \\ 1/8, & \text{if } 1 \leq t < 3 \\ 3/8, & \text{if } 3 \leq t < 5 \\ 7/8, & \text{if } 5 \leq t < 7 \\ 1, & \text{if } t \geq 7 \end{cases}$$

- (a) Make a nice plot of this function using python in a Jupyter notebook.
- (b) The points of discontinuity or jumps in the graph are the possible values of T , and the height of the jumps are the corresponding probabilities. Write down the probability distribution table for T with the possible values, and the corresponding probabilities.
- (c) Find $P(1.4 < T < 6)$
- (d) Compute the conditional probability $P(T \leq 5 | T \geq 2)$.

2. The data “student-por.csv” consists of Portuguese language course students final grades (column “G3”) in two schools “GP” (Gabriel Pereira) and “MS” (Mousinho da Silveira). Let X and Y stand for the grade “G3” in the two schools respectively. Note that X and Y are discrete variables taking possible values $0, 1, 2, \dots, 20$. [15 points]

- (a) From the data find the probability distributions of X and Y . Plot the two distributions separately and also side-by-side.
- (b) Compute the expected (means) values $E(X)$, $E(Y)$ and the standard deviations $SD(X)$ and $SD(Y)$ from the probability distributions.
- (c) Compare the performance of students in the two schools using the plots, the means and standard deviations.

Please do this Problem 2 in a Jupyter notebook.

3. A private pilot wishes to insure his airplane for \$ 400,000. The insurance company estimates that a total loss will occur with probability 0.002, a 50% loss with probability 0.01, and a 25% loss with probability 0.1. Ignoring all other partial losses, what premium should the insurance company charge for each year to realize an average profit of \$ 500. [5 points]